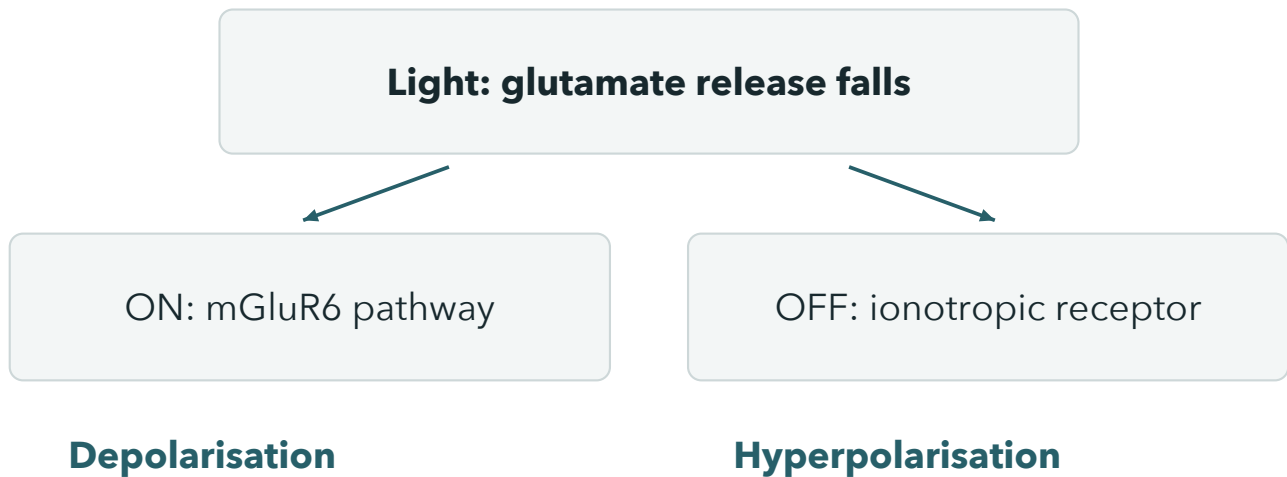


Retinal circuits and contrast coding

Part 1 | One transmitter, opposite bipolar responses

The receptor and intracellular pathway determine response polarity.



Less photoreceptor glutamate during illumination has opposite effects on ON and OFF bipolar pathways.

Locate the sign inversion

ON bipolar cells use the mGluR6 signalling pathway linked to TRPM1; OFF pathways use ionotropic glutamate receptors. The same decrease in transmitter can therefore depolarise one pathway and hyperpolarise the other. State the presynaptic change before predicting the postsynaptic voltage.

A response polarity experiment

Receptor-specific manipulations distinguish release failure from postsynaptic signalling failure.

Manipulation

Feature	Preserved process	Affected inference
Photoreceptor release held fixed	Presynaptic output	Receptor pathway can be tested
ON pathway selectively impaired	OFF signalling may remain	Not global photoreceptor silence
Reduced photoreceptor output	Both receive altered input	Downstream effects can diverge

Design the comparison

Recording only one bipolar class cannot establish that glutamate release is absent. Pair a presynaptic measurement with both response classes, or restore one receptor pathway selectively. A common input and different outputs provide a stronger test of sign inversion than a single voltage trace.

Centre and surround oppose each other

Spatial weighting makes local boundaries more informative than uniform illumination.

ON-centre example

Stimulus	Firing tendency
Small bright centre	Centre excitation
Bright surrounding annulus	Surround inhibition
Uniform centre and surround	Partial cancellation

Read the geometry

Keep wavelength and temporal conditions fixed when testing spatial antagonism.

The annulus must spare the centre to isolate the surround.

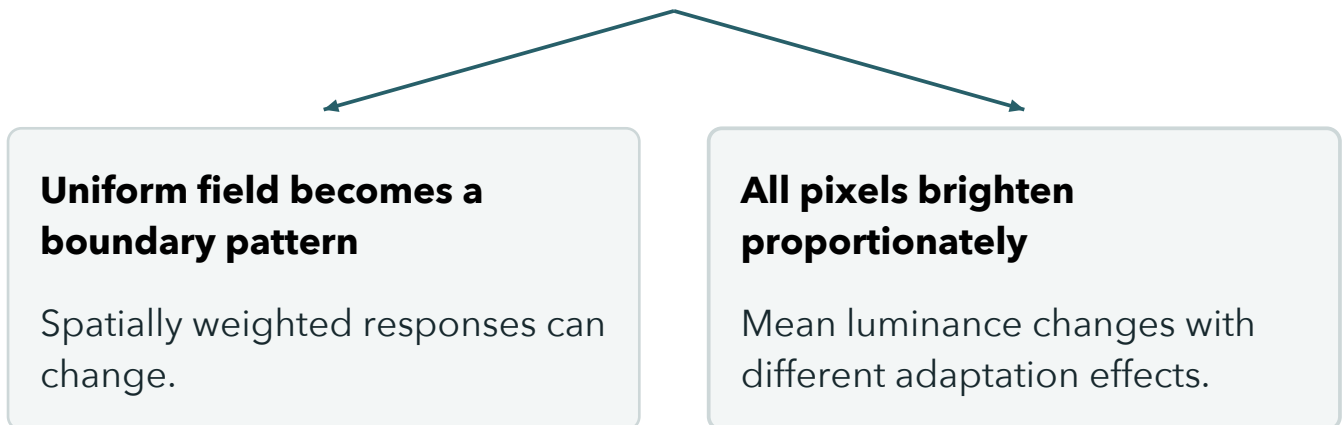
Antagonism does not uniquely identify a specialised named cell subtype.

Contrast and total luminance are distinct

Two scenes can have the same mean luminance but different spatial information.

Compare matched scenes

What changes?



An original thought experiment

Compare a grey field with a fine bright-dark pattern adjusted to the same average luminance. A neuron that responds differently is not simply summing all incident photons. Repeat at several spatial scales to separate contrast sensitivity from a preferred boundary location.

Circuit claims require controlled evidence

A receptive-field description is not a complete cellular diagnosis.

Evidence

Feature	Supported conclusion	Unsupported leap
Opposing centre/surround	Spatial antagonism	Unique ganglion subtype
ON/OFF dissociation	Different response pathways	One specific causal gene
Normal gross anatomy	Structure preserved at measured scale	Normal physiological signalling

Examination approach

Name the measured level: transmitter release, bipolar voltage or ganglion firing. Then ask whether the proposed mechanism acts upstream, within that level or downstream. This prevents a cortical explanation being used for an experimentally isolated retinal response.